Research Statement

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Graph mining plays a fundamental role in many real-world domains, such as healthcare, financial security and social science. Years of research in this area have developed a plethora of theories, algorithms and systems that are successful in answering *what/who* types of questions. For example, *what* are the most relevant web pages with respect to a given query? *Who* can be grouped into the same community? Despite the prosperity of graph[1] mining, a fundamental question largely remains nascent: how can we make graph mining algorithms *trustworthy*? To answer this question, it is crucial to propose a paradigm shift for graph mining, from answering *who* and *what* to answering *how* and *why*. And trustworthy graph mining could be beneficial in a variety of real-world scenarios, including financial fraud detection, personalized medicine and disaster risk management. In order to achieve the *trustworthy graph mining*, several key pillars of graph mining algorithms need to be explored. The first pillar refers to transparency, which focuses on demonstrating how the mining results of a given network mining algorithm relate to the underlying graph structure and explaining why a graph mining algorithms make a certain decision. The second pillar refers to fairness, which explores methods to mitigate potential biases in graph mining process or results. The third pillar refers to robustness, which focuses on making the graph mining process resilient to noises or adversarial activity. The last pillar refers to privacy, which studies how to prevent private data leakage. My research aims to develop computational models, algorithms and tools underpinning all four pillars, by focusing on the following research problems. An overview of my research is presented in Figure 1.

Thrust 1: Transparent Graph Mining. State-of-the-art graph mining algorithms almost exclusively focuses on answering *what/who* types of questions accurately, while sparse efforts are made to enhance the transparency of graph mining algorithms. Consequently, it fails to answer questions like *how the results of a graph mining algorithm relate to the underlying graph structure*. Moreover, end users, who are often not experts in data mining and machine learning, have little understanding on how a particular decision is made by the graph mining algorithm and do not trust the algorithm. A transparent graph mining algorithm not only increases trust of end users, but also helps us understand in which scenarios the algorithm would fail.

Thrust 2: Fair Graph Mining. The ubiquity of graphs in real world has nourished the developments of numerous computational models. However, many widely-used graph mining algorithms suffer from potential discrimination against certain individual or group of individuals. Ensuring fairness on graphs is challenging due to two key challenges: (1) theoretical challenge on the implication of non-IID graph data in bias quantification and mitigation; (2) algorithmic challenge on the dilemma of balancing model accuracy and fairness. Therefore, it is crucial to study (1) how to mitigate the bias in an *existing* graph mining algorithms and (2) how to design *new* fairness-aware graph mining algorithms to alleviate the accuracy-fairness trade-off.

Thrust 3: Robust Graph Mining. Recent research has demonstrated that graph mining algorithms could be sensitive to noisy and uncertain nodes/edges as well as being vulnerable to adversarial attacks. The unique structure of graph data (i.e., the inter-connections among nodes) brings unique challenge in studying robust graph mining algorithm. For example, a malicious agent could affect the model prediction on a particular node without any direct operation on that node. Thus, it is necessary to learn robustified graphs and/or develop robust graph mining models that are resilient to noises, uncertainties and malicious users while maintaining high utility. Robust graph mining is the key to answering questions like: *how* can we alleviate the traffic congestion when unusual events (e.g., car accidents) happens? *How* do we ensure recommendation quality when there are fake ratings in the system?

Thrust 4: Private Graph Mining. The advent of recent legislation (e.g., General Data Protection Regulation) raises the increasing concerns on protecting the sensitive and private information on graph mining algorithms. As a key cornerstone in trustworthy graph mining, private graph mining asks for additional requirements on privacy protection due to the interconnections among nodes, i.e., the sensitive information of a node may contain not only its own demographic information but also its relationships to other nodes in the graph. Regarding private graph mining, it is essential to understand *why* a graph mining algorithm would leak private information and *how* to ensure that the sensitive information of a node would not be deliberately utilized by graph mining algorithms so as to protect the privacy in graph mining algorithms.

[1] We use the terms 'network' and 'graph' interchangeably.

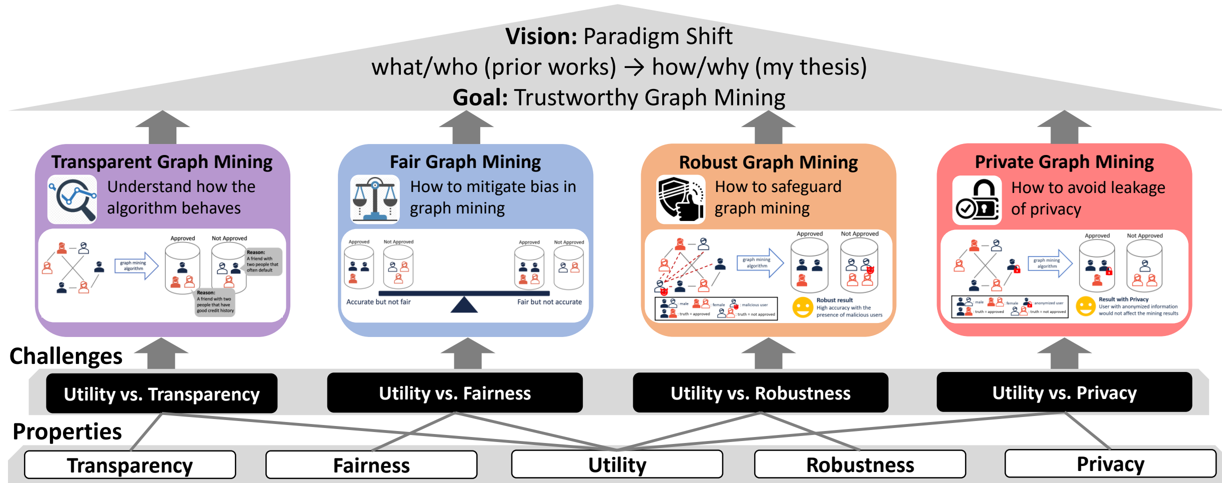

Figure 1: An overview of my research on trustworthy graph mining.

1 Current Achievements

My research contributions so far have primarily focused on transparent graph mining, fair graph mining and robust graph mining. In general, my current progress lies in auditing graph mining algorithms from the aspect of transparent graph mining, enforcing individual fairness, degree-related fairness and group fairness from the aspect of fair graph mining and quantifying the uncertainty on graph neural networks from the aspect of robust graph mining.

1.1 Transparent Graph Mining

Despite tremendous progress, graph mining algorithms are often opaque with respect to *how the results of a graph mining algorithm relate to the underlying graph structure*. In my research, we first start from auditing PageRank algorithm and then present a generic algorithmic framework to audit the graph mining algorithm.
Auditing PageRank. PageRank is one of the most successful ranking algorithms on graphs due to its mathematical elegance and superior performance. Important as it might be, end-users often do not understand how the ranking results were derived as it lacks intuitive way to explain the ranking results. To solve this problem, we propose to identify *which* nodes are the most important ones in order to understand *why* the ranking algorithm gives a particular ranking result, which is the family of Aurora algorithms. The key idea of Aurora is to measure the influence of each graph element as the rate of change (i.e., derivative) of a scalar-valued function over the PageRank vector. By our definition of influence, we demonstrate that Aurora can effectively identify influential graph elements with an $(1 - 1/e)$ approximation ratio due to the diminishing returns property of influence. Empirical experimental results show that Aurora can effectively identify influential graph elements compared with several heuristic baseline methods.
Network Derivative Mining. We further generalize the key insights we developed in Aurora and develop a general algorithmic framework to audit a variety of graph mining models, which is referred to as the network derivative mining problem (N2N). It aims to construct a derivative network, each of whose edges quantitatively measure the influence as the rate of change of a function over the mining results induced by the given network mining algorithm. We instantiate the proposed N2N framework with a variety of graph mining algorithms, including HITS ranking, spectral clustering and matrix completion. For each instantiation, we develop efficient solution to construct the derivative network with linear complexity in both time and space. Empirical results show that our proposed N2N framework outperforms all heuristic and state-of-the-art methods in the task of attacking the given network mining algorithm.
Impact/Results. We are the first to study PageRank auditing problem [7] and further develop it into an online demonstration system [2]. The long version of this study has been published in IEEE TKDE [11], which is the premium journal in the field of data mining. Our proposed network derivative mining [4] is a

new mining paradigm and could potentially impact many research areas, including explainable graph mining, adversarial learning on graphs, optimal graph learning and counterfactual learning on graphs. My research on truncated PageRank was successfully adopted in real-world products at Meta to benefit millions of users.

1.2 Fair Graph Mining

Though graph mining algorithms achieve state-of-the-art performance in many downstream tasks, many questions in relation to the algorithmic fairness are unanswered, e.g., *are the graph mining results fair? If not, how to mitigate bias? How would imposing fairness constraints impact the mining performance (e.g., classification accuracy)?* To answer these questions, we systematically study fairness on graph mining.

Individual Fairness. The sparse literature on fair graph mining has almost exclusively focused on group fairness. However, individual fairness, which promises fairness in a finer node-level granularity, has not been well studied. In my research, we study the problem of individual fairness on graph mining (InFoRM). We first quantitatively measure the individual bias as the trace of a quadratic form of the mining results. And we propose a family of three mutually complementary algorithms, namely DEBIASING THE INPUT GRAPH, DEBIASING THE MINING MODEL and DEBIASING THE MINING RESULTS, to mitigate our proposed individual bias measure. Each method is formulated as an optimization problem, using effective and efficient solvers, which can be applied to a variety of graph mining algorithms. Furthermore, we conduct a thorough analysis to understand the cost of enforcing individual fairness (i.e., the difference between the graph mining results with and without fairness considerations). We prove that the cost of model-agnostic DEBIASING THE MINING RESULTS method relates to several factors of the graph structure: the number of nodes, the difference between the adjacency matrix and the node-node similarity matrix, the rank and the largest singular value of the adjacency matrix. To generalize InFoRM from clustering on single network to clustering on multi-view network, we further propose iFiG, which is an efficient solver to ensure individual fairness on multi-view graph mining by solving a multi-objective optimization problem. Empirical analysis demonstrates that the proposed iFiG method is the best one to ensure individual fairness on multi-view networks without much sacrifice on the performance of graph mining algorithms. Moreover, to avoid the distance comparison in Lipschitz inequality-based definition of individual fairness (e.g., InFoRM), we define the ranking-based individual fairness. Based on that, we propose a generic plug-and-play framework named REDRESS to promote the ranking-based individual fairness for any graph neural networks. Our proposed method can effectively enforce ranking-based individual fairness while maintaining the model utility.

Degree-related Fairness. Despite the success of Graph Convolutional Network (GCN) in many real-world applications, it often exhibits performance disparity with respect to node degrees, resulting in worse predictive accuracy for low-degree nodes. In my research, we propose RawlsGCN to mitigate the degree-related performance disparity by enforcing the Rawlsian difference principle, which is originated from the theory of distributive justice. My research has revealed the root cause of this degree-related unfairness by analyzing the gradients of weight matrices. Then we normalize the influence of each node on the gradients through doubly stochastic normalization on the input graph. RawlsGCN is able to mitigate the bias without modification on the GCN architecture or introduction of additional parameters. Empirical evaluations on real-world graphs demonstrate the effectiveness and efficiency of RawlsGCN.

Group Fairness. Group fairness is a fundamental fairness notion, which aims to enforce the parity of a certain statistical measure among the pre-defined demographic groups. Many existing works in group fairness focuses on debiasing with respect to a single sensitive attribute, despite the fact that co-existence of multiple sensitive attributes is commonplace (e.g., gender, race, marital status). In addition, existing works often optimize a surrogate function of the statistical correlation between the sensitive attribute and the model prediction. In my research, we propose InfoFair, which directly enforces statistical parity on multiple sensitive attributes simultaneously by minimizing the mutual information between the model prediction and the joint variable of all interested sensitive attributes. To minimize the mutual information between two high-dimensional random variables, we leverage a variational representation of mutual information. The key steps to optimize the variational representation include (1) utilizing a decoder to 'reconstruct' the sensitive attributes using the model prediction and (2) modeling the density ratio computation as a class probability estimation problem. Experiments on real-world data demonstrate that InfoFair achieves the best trade-off between fairness and accuracy compared with the baseline methods. Moreover, our proposed InfoFair is generalizable to settings beyond the scope of fair classification with categorical sensitive attributes in the majority of existing works.

Human-in-the-loop Visual Analytics for Fair Graph Mining. In my collaboration with researchers from Arizona State University, we propose FairRankVis, an interactive visual analytics framework for exploring fairness in graph ranking. An overview of its interface is shown in Figure 2. The framework supports exploration of human-guided fairness definitions (i.e., group fairness, individual fairness) with respect to a human-defined combination of sensitive attributes and enables to diagnose the model trade-offs under different fairness definitions.

Impact/Results. We are the first to systematically study the problem of individual fairness on graph mining, which has led to three publications in KDD 2020 [3], KDD 2021 [1] and IEEE Big Data 2022 [12], all of which are top data mining conferences. Our proposed RawlsGCN that enforces Rawalsian difference principle is published in WWW 2022, which is a premium conference on Web [10]. Our proposed InfoFair that minimizes a variational representation of mutual information is published as at IEEE Big Data 2022 [8]. The visual analytics framework FairRankVis is published in IEEE Transactions on Visualization and Computer Graphics, a premium journal for visualization [13]. We have also held two lecture-style tutorials at CIKM 2021 and KDD 2022, both of which are top conferences in data mining [5, 6]. My works along this line have been the key deliverables to an NSF project (Fairness in AI program) and the key component in the graph-based recommender system for Reels (i.e., a short-form video platform by Meta). I was invited to give two guest lectures at Virginia Tech and Pennsylvania State University, respectively, a keynote talk at the EAI-KDD workshop at KDD 2022 and a lecture-style tutorial at SDM 2023 on this topic.

Figure 2: The user interface of FairRankVis.

1.3 Robust Graph Mining

Robustness characterizes the ability of graph mining in tolerating adversarial attacks or natural variations (e.g., noise, uncertainty) on graphs. Compared with rich studies in developing robust graph mining algorithms against adversarial activities, it remains nascent on the fundamental question like *how* uncertain is a graph mining algorithm in its mining results? To answer this question, we study the post-hoc uncertainty quantification on graph neural networks.

Uncertainty Quantification on Graph Neural Networks. The vast majority of existing works on Graph Convolutional Network (GCN) primarily focus on the accuracy while ignoring how confident or uncertain a GCN is with respect to its predictions. The scarce existing efforts either fail to provide deterministic quantification or have to change the training procedure of GCN. In my research, we propose JuryGCN, which is the first frequentist-based approach to quantify the uncertainty of GCN deterministically without modifying the GCN architecture or introducing additional parameters. Its key idea is to quantify the uncertainty of a node as the width of confidence interval by a jackknife estimator. And the influence function is leveraged to estimate the change in GCN parameters without re-training to scale up the computation. Extensive evaluations in the tasks of both active learning and semi-supervised node classification demonstrate the efficacy of JuryGCN.

Impact/Results. We are the first to propose the frequentist-based uncertainty quantification approach on graph neural networks. The work has been published in the premium data mining conference KDD 2022 with both oral and poster presentation [9]. This work was part of the deliverables to an NSF project (MoDL SCALE program) and is being further exploited in an ongoing DARPA project (INCAS).

2 Future Research Directions

While automated decision-making techniques significantly benefit our society, there is rising societal awareness that calls for the trustworthiness of these techniques. My current contributions have laid solid foundation in the transparency, fairness and robustness of graph mining algorithms, which prepares me well for further investigating the trustworthiness of artificial intelligence (AI) in the future. In the short term (the first 3-5 years), I will mainly continue investigating the trustworthiness of graph mining underpinning all four pillars (i.e., transparent, fair, robust and private graph mining). In the long term, I am interested in fundamental theories behind the interconnections among different aspects of trustworthy AI and principled solutions to deploy graph mining in an open world, which is a more realistic but even more challenging setting. In the meantime, I will also actively collaborate with researchers from different domains to conduct interdisciplinary research.

2.1 Short-Term Plans

Short-Term Plan #1: Safeguarding Fair Graph Mining. Existing fair graph mining models aims to find the best way to mine the input graph and ensure fairness. It is not surprising that the real-world data is often noisy and could contain adversarial activities. As such, for deployable fair graph mining, it is crucial to safeguard fair graph mining by studying *how* robust it is against adversaries and distributional shifts.

Regarding adversarial robustness, the graph may be contaminated by malicious users who deliberately establish connections with other nodes from certain demographic groups, so as to mislead the fair graph mining algorithm to discriminate the underrepresented group (i.e., fairness attacks). Thus, simply training a fair graph mining algorithm without consideration of fairness attacks would fail to mitigate the bias and/or behave in a maximally biased way. To mitigate the risk of fairness attacks, we will investigate (1) effective approaches to achieve the fairness attacks and (2) effective methods to robustify the fair graph mining algorithms against fairness attacks. A potential solution to both problems would be leveraging multi-level optimization techniques where different levels optimizes different objectives to achieve fair attacks (e.g., maximizing the bias measure) or robustify fair graph mining algorithms (e.g., imputing the input graph to defend against fairness attacks).

In terms of robustness against distributional shifts, existing fair graph mining algorithms rely on learning debiased representations with observed labels, while overlooking the uncertainty of the observations. The uncertainty brings several challenges in fair graph mining. First, uncertain observations could result in a covariate shift between distributions of the training set and the test set, which might violate the fundamental assumption about the same distribution between training data and test data in graph neural networks. Second, the uncertainty could mislead the fair mining algorithm toward sub-optimal training directions, resulting in unfairness in the testing phase. To tackle these challenges, I will devote myself to developing effective approaches to achieve fairness on graphs under uncertainty by calibrating the fair graph mining algorithms with the uncertainty of each node/graph.

Potential Funding Sources: NSF CAREER program, ONR Science of Artificial Intelligence program, Amazon research award.

Short-Term Plan #2: Explaining Graph Mining beyond Utility. To date, the majority of explainable graph mining has been solely focusing on explaining *why* a graph mining algorithm makes the 'best' decision for its utility. However, with the growing concerns on the trustworthiness of graph mining algorithms, users may also wish to be informed if the decision is fair regardless of his/her demographic information or whether the decision is made by using his/her private information. To accommodate the increasing need over the transparency of graph mining algorithms, it is important to understand *how* to explain graph mining algorithms from different social aspects.

To this end, I plan to develop computational algorithms and systems to explain graph mining algorithm beyond its utility. I believe the study of explainability beyond utility provides significant contributions to understand different social aspects of graph mining algorithms as well. For example, by explaining the fairness of graph mining algorithms, it enables the ability to discover the graph elements that have largest impact on the bias/fairness of the graph mining algorithm. With the knowledge on these graph elements, it could (1) provide us fundamental understanding on why a graph mining algorithm makes a biased decision and/or (2) open the door to develop fair graph mining algorithms via perturbations on such graph elements.

By explaining the robustness of graph mining algorithms, it could identify which nodes are noisy or malicious with respect to the graph mining task, which may in turn help certify the robustness of graph mining.
Potential Funding Sources: NSF III program, NSF RI program, DARPA AIE program.
Short-Term Plan #3: Unlearning on Graphs with Theoretical Guarantees. There has been growing attention on the privacy protection of automated decision-making algorithms. Consequently, multiple jurisdictions have implemented the 'right to be forgotten', which empowers a user the right to request any organizations to delete his/her own private data in a rigorous manner. In the context of graph mining, the user could request an administrator of a graph mining algorithm to remove the effect of his/her private data on the graph mining algorithm that is previously trained with his/her private data. A naive solution to achieve such removal is to simply re-train the graph mining algorithm without the corresponding data record, which is either computationally prohibitive due to large scale of real-world data or infeasible due to unavailable data records that are previously used for training (e.g., out of the retention cycle of the data storage). Sparse efforts have been made to study efficient unlearning techniques on graph mining models, which either fail to adapt to real-world deployed graph mining algorithms or lack of theoretical understanding on to what extent the private information has been removed. To fill the gap between real-world graph mining models and certified machine unlearning on non-graph data, I seek to study unlearning techniques for graph mining algorithms with rigorous theoretical understanding and strong empirical performance. To unlearn for graph mining algorithms in the real world, I plan to study machine unlearning on nonlinear graph neural networks with the development of efficient unlearning techniques on linear graph neural networks as a preliminary. Besides, I will establish theories and metrics to analyze the theoretical guarantees on the (im-)possiblity of machine learning on graph mining algorithms.
Potential Funding Sources: NSF SaTC program, DARPA CSL program, Meta grant.

2.2 Long-Term Plans

Long-Term Plan #1: Graph Mining in an Open World. Current graph mining algorithms are primarily developed and evaluated in a closed or domain-specific environment. However, the real world is often complicated and keeps evolving. As a consequence, existing graph mining algorithms could be vulnerable to the real world with noises, distributional shifts, new users and unobserved domains. For my long-term future research, I would like to study graph mining in an *open world environment*. Compared with graph mining in a domain-specific setting, graph mining in an open world should (1) maintain its trustworthiness by being resilient to bias, noise, distributional shift and adversarial activities and (2) be capable of accommodating the incoming new users and domains that do not exist in the history. To adapt graph mining in an open world, I believe it relies on (1) a domain-invariant learning framework which enables the knowledge to be transferred across domains and (2) a lifelong learning framework which continually learns knowledge in the new domain without forgetting knowledge in the old domains. Though being challenging, a solution to open-world graph mining can enhance the trustworthiness of graph mining in a complicated environment and facilitate the deployment of graph mining algorithms to benefit human-being in the long run.
Potential Funding Sources: DARPA ECOLE program, ARO BAA for Fundamental Research.
Long-Term Plan #2: Interconnections among Aspects of Trustworthy AI. To date, major efforts in trustworthy AI aim to guarantee one aspect of trustworthy AI (e.g., transparency, fairness, robustness, privacy), while overlooking the interrelationships among them. Without deep understanding on their implicit relationships, the trustworthy AI techniques would not be trustworthy and conscious enough to adapt the learned knowledge on unknown factors that do not exist previously. Thus, in the long term, I am devoted to studying the interconnections among different aspects of trustworthy AI, which helps make an AI algorithm to be trustworthy in general rather than respecting to only one aspect. For example, a robust AI algorithm may help remove spurious correlation in model explanations and also ensure privacy; an explainable AI algorithm may help identify if there exists privacy leakage and if the algorithm is making biased decision. I am also interested in exploring the potential applications of such general trustworthy AI such as medical diagnosis and policy making.
Potential Funding Sources: NSF SaTC program, DARPA I2O program.
Interdisciplinary Research. Multi-disciplinary collaboration is one of the major driving forces behind many scientific studies. Throughout my academic career, I am privileged to have several enjoyable collaborations with researchers from different areas in computer science (e.g., human-computer interaction) [7, 2,

11, 3, 13, 8]. These collaborations benefit me in not only broadening my vision and knowledge but also enhancing my research impact. Based on my research background on graph mining and trustworthy artificial intelligence, I look forward to working with researchers from different areas of computer science (e.g., human-computer interaction, control systems, theoretical computer science, etc.) and different disciplines (e.g., finance, healthcare, social science, anthropology, psychology, etc.) for trustworthy graph mining.

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